

Superstore Data Analysis Python Project

Objective:

Analyze sales, profit, customer behavior, discount impact, and operational performance using Python.

Tools Used:

- Python
- Pandas
- Matplotlib
- Seaborn

Dataset:

Sample Superstore Dataset

Business Problem

The Superstore business wants to understand its sales performance, profitability, customer behavior, and operational efficiency. The company is facing uneven profits across regions and products, and management needs data-driven insights to improve business performance. Main Objectives: Identify high-performing and loss-making areas Analyze sales and profit trends Understand the impact of discounts on profit Find best-selling and least profitable products Improve business decision-making using data analysis

Importing Required Libraries

```
In [3]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Reading the CSV File

```
In [4]: data = pd.read_csv(r"C:\superstoore\Sample - Superstore.csv", encoding="latin1")
```

Important Columns: Order ID Order Date Customer Name Region State Category Sub-Category Sales Profit Quantity
Discount Ship Mode Tools Used:

.head()

```
In [6]: data.head(4)
```

Out[6]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Co
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	L
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	L
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	L
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	L

4 rows × 21 columns

Data Cleaning

In [21]:

data.info()

```

<class 'pandas.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row ID                 9994 non-null   int64
1   Order ID               9994 non-null   str
2   Order Date             9994 non-null   str
3   Ship Date              9994 non-null   str
4   Ship Mode              9994 non-null   str
5   Customer ID            9994 non-null   str
6   Customer Name          9994 non-null   str
7   Segment                9994 non-null   str
8   Country                9994 non-null   str
9   City                   9994 non-null   str
10  State                  9994 non-null   str
11  Postal Code            9994 non-null   int64
12  Region                 9994 non-null   str
13  Product ID             9994 non-null   str
14  Category                9994 non-null   str
15  Sub-Category           9994 non-null   str
16  Product Name           9994 non-null   str
17  Sales                   9994 non-null   float64
18  Quantity                9994 non-null   int64
19  Discount                9994 non-null   float64
20  Profit                  9994 non-null   float64
dtypes: float64(3), int64(3), str(15)
memory usage: 1.6 MB

```

In [22]: `data.describe()`

Out[22]:

	Row ID	Postal Code	Sales	Quantity	Discount	Profit
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203	28.656896
std	2885.163629	32063.693350	623.245101	2.225110	0.206452	234.260108
min	1.000000	1040.000000	0.444000	1.000000	0.000000	-6599.978000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000	1.728750
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000	8.666500
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000	29.364000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000	8399.976000

In [9]: `data.columns`

Out[9]: Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',
'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category',
'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit'],
dtype='str')

In [10]: `data.shape`

```
Out[10]: (9994, 21)
```

```
In [12]: data.count()
```

```
Out[12]: Row ID          9994
Order ID          9994
Order Date        9994
Ship Date         9994
Ship Mode         9994
Customer ID       9994
Customer Name     9994
Segment          9994
Country           9994
City              9994
State             9994
Postal Code       9994
Region            9994
Product ID        9994
Category          9994
Sub-Category      9994
Product Name      9994
Sales             9994
Quantity          9994
Discount          9994
Profit            9994
dtype: int64
```

```
In [26]: data.isnull().sum()
```

```
Out[26]: Row ID          0
Order ID          0
Order Date        0
Ship Date         0
Ship Mode         0
Customer ID       0
Customer Name     0
Segment          0
Country           0
City              0
State             0
Postal Code       0
Region            0
Product ID        0
Category          0
Sub-Category      0
Product Name      0
Sales             0
Quantity          0
Discount          0
Profit            0
dtype: int64
```

```
In [24]: print("Duplicate Rows :", data.duplicated().sum())
```

```
Duplicate Rows : 0
```

- How many unique customers, products, and orders are there?

```
In [33]: data[['Customer ID', 'Product ID', 'Order ID']].nunique()
```

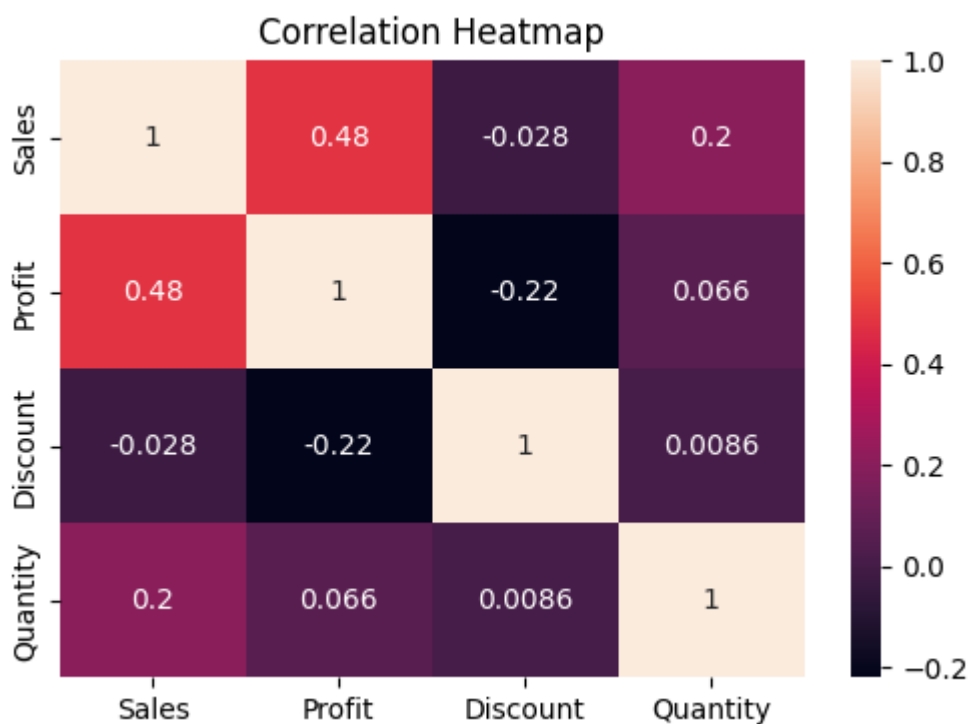
```
Out[33]: Customer ID      793
Product ID    1862
Order ID      5009
dtype: int64
```

```
In [20]: plt.figure(figsize=(6,4))

sns.heatmap(
    data[['Sales', 'Profit', 'Discount', 'Quantity']].corr(),
    annot=True
)

plt.title("Correlation Heatmap")

plt.show()
```



Sales and Profit have a positive relationship. Discount has a negative correlation with Profit, showing that higher discounts reduce profitability.

Sales & Profit Analysis

• What is the total sales, profit, and quantity?

```
In [37]: totals = data[['Sales', 'Profit', 'Quantity']].sum().round(2)

print("Total Sales :", totals['Sales'])
print("Total Profit :", totals['Profit'])
print("Total Quantity :", totals['Quantity'])
```

```
Total Sales : 2297200.86
Total Profit : 286397.02
Total Quantity : 37873.0
```

• Which region generates the highest sales?

```
In [13]: plt.figure(figsize=(7,4))

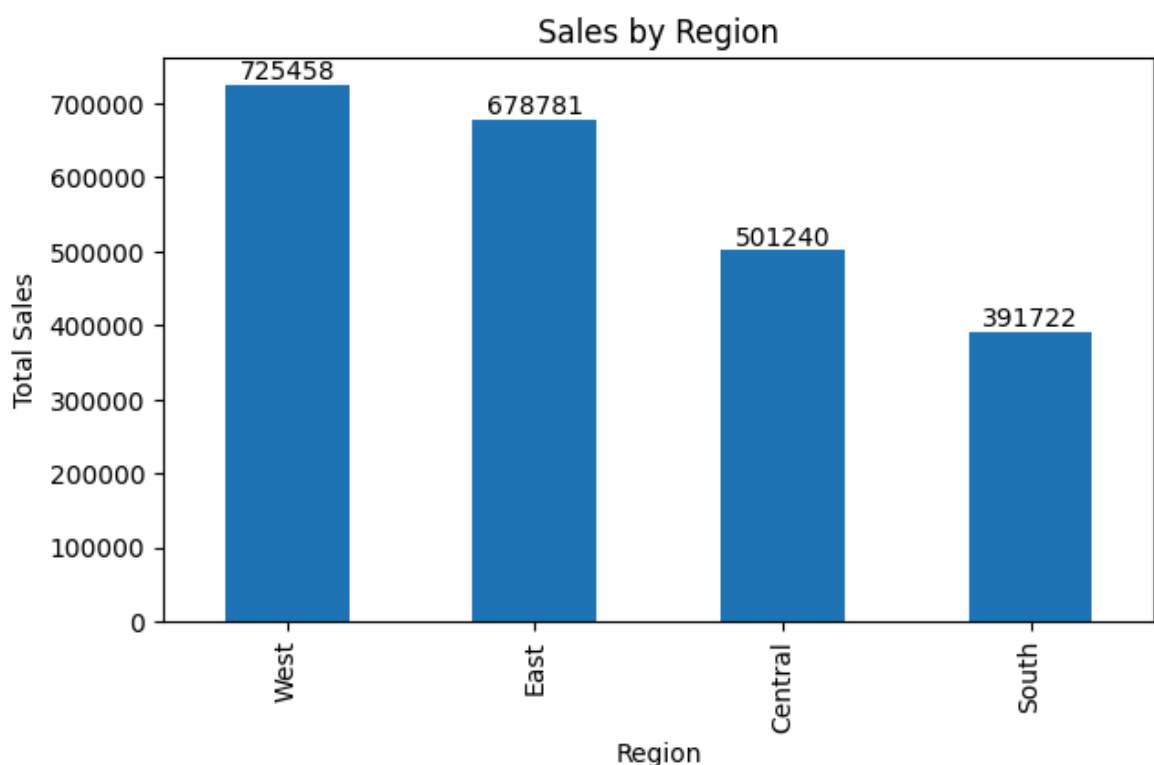
region_sales = data.groupby('Region')['Sales'].sum().sort_values(ascending=False)

ax = region_sales.plot(kind='bar')

plt.bar_label(ax.containers[0], fmt='%.0f')

plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")

plt.show()
```



• Which city gives maximum sales?

```
In [125... max_city_sale = data.groupby('City')['Sales'].sum()

print('Maximum Sales City :', max_city_sale.idxmax())
print('Maximum Sales Amount :', max_city_sale.max())
```

Maximum Sales City : New York City

Maximum Sales Amount : 256368.161

• Which state is most profitable / loss-making?

```
In [123... state_profit = data.groupby('State')['Profit'].sum()

print("Most Profitable State :", state_profit.idxmax())
```

```
print("Highest Profit Amount :", round(state_profit.max(), 2))

print("Most Loss-Making State :", state_profit.idxmin())
print("Lowest Profit Amount :", round(state_profit.min(), 2))
```

Most Profitable State : California
 Highest Profit Amount : 76381.39
 Most Loss-Making State : Texas
 Lowest Profit Amount : -25729.36

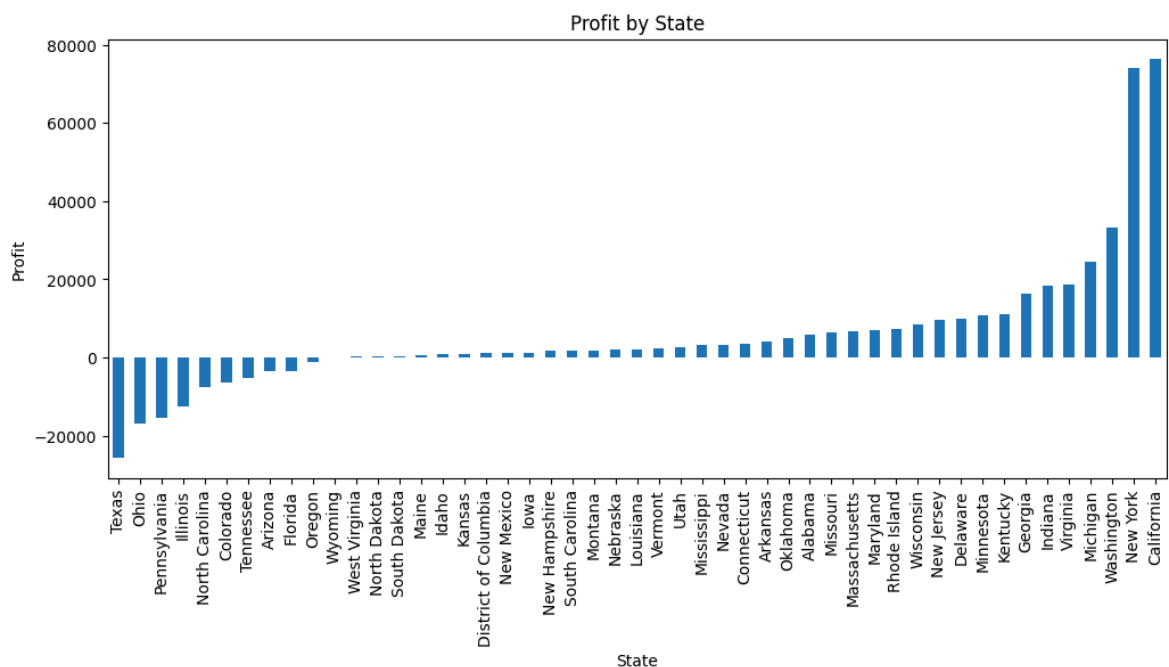
```
In [14]: plt.figure(figsize=(12,5))

state_profit = data.groupby('State')['Profit'].sum().sort_values()

state_profit.plot(kind='bar')

plt.title("Profit by State")
plt.xlabel("State")
plt.ylabel("Profit")

plt.show()
```



Profit Margin Analysis

```
In [25]: data['Profit Margin %'] = (data['Profit'] / data['Sales']) * 100

margin = data['Profit Margin %'].mean()

print("Average Profit Margin :", round(margin, 2))
```

Average Profit Margin : 12.03

Category & Sub-Category Analysis

- Which category (Furniture, Office Supplies, Technology) has highest sales?

```
In [23]: data.groupby('Category')['Sales'].sum().sort_values(ascending=False)
```

```
Out[23]: Category
Technology      836154.0330
Furniture       741999.7953
Office Supplies  719047.0320
Name: Sales, dtype: float64
```

```
In [15]: plt.figure(figsize=(6,4))

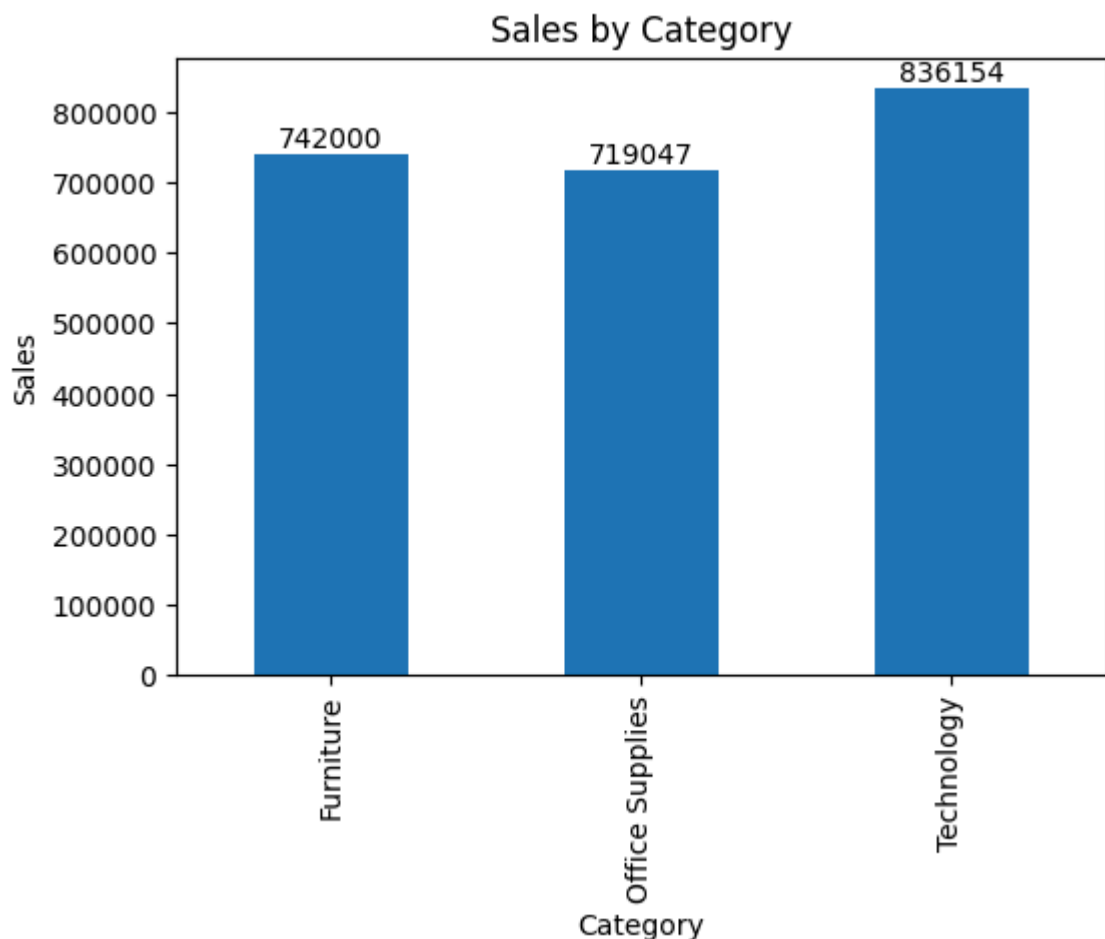
category_sales = data.groupby('Category')['Sales'].sum()

ax = category_sales.plot(kind='bar')

plt.bar_label(ax.containers[0], fmt='%.0f')

plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")

plt.show()
```



• Which sub-category is most sold?

Sub_category

```
In [166... data.groupby('Sub-Category')[['Quantity', 'Sales']].sum().sort_values(by='Quanti
```

Out[166...

	Quantity	Sales
Sub-Category		
Binders	5974	203412.7330
Paper	5178	78479.2060
Furnishings	3563	91705.1640
Phones	3289	330007.0540
Storage	3158	223843.6080
Art	3000	27118.7920
Accessories	2976	167380.3180
Chairs	2356	328449.1030
Appliances	1729	107532.1610
Labels	1400	12486.3120
Tables	1241	206965.5320
Fasteners	914	3024.2800
Envelopes	906	16476.4020
Bookcases	868	114879.9963
Supplies	647	46673.5380
Machines	440	189238.6310
Copiers	234	149528.0300

Most Sold Out Sub_category

```
In [137... category = data.groupby('Sub-Category')['Sales'].sum()

print('Highest Profit Sub-Category :', (category.idxmax()))
print('Profit Amount :', round(category.max(),2))
```

Highest Profit Sub-Category : Phones

Profit Amount : 330007.05

The Phones sub-category generates the highest sales of approximately 330007.05.

• Which sub-category is most loss-making?

```
In [150... category = data.groupby('Sub-Category')['Profit'].sum()

print('Most Loss-Making Sub-Category :', category.idxmin())
print('Lowest Profit Amount :', round(category.min(), 2))
```

Most Loss-Making Sub-Category : Tables

Lowest Profit Amount : -17725.48

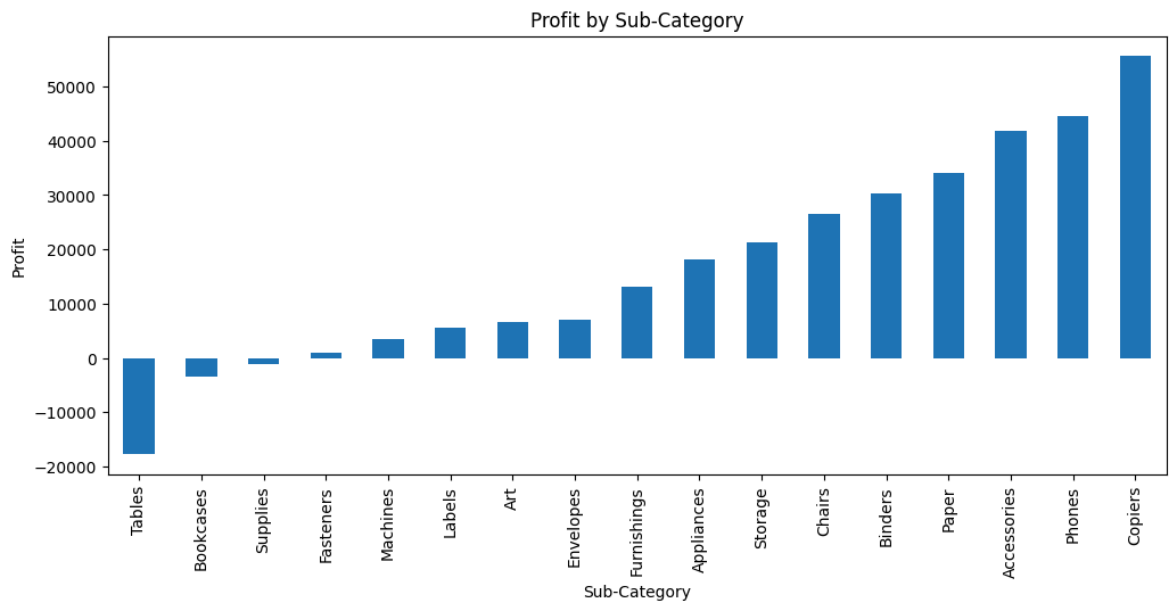
```
In [16]: plt.figure(figsize=(12,5))

sub_profit = data.groupby('Sub-Category')['Profit'].sum().sort_values()
```

```
sub_profit.plot(kind='bar')

plt.title("Profit by Sub-Category")
plt.xlabel("Sub-Category")
plt.ylabel("Profit")

plt.show()
```



• Which sub-category has highest discount impact?

Sub-category Discount and profit

```
In [163... data.groupby('Sub-Category')[['Discount', 'Profit']].sum().sort_values(by='Disco
```

Out[163...

	Discount	Profit
Sub-Category		
Binders	567.00	30221.7633
Phones	137.40	44515.7306
Furnishings	132.40	13059.1436
Chairs	105.00	26590.1663
Paper	102.60	34053.5693
Tables	83.35	-17725.4811
Appliances	77.60	18138.0054
Storage	63.20	21278.8264
Accessories	60.80	41936.6357
Art	59.60	6527.7870
Bookcases	48.14	-3472.5560
Machines	35.20	3384.7569
Labels	25.00	5546.2540
Envelopes	20.40	6964.1767
Fasteners	17.80	949.5182
Supplies	14.60	-1189.0995
Copiers	11.00	55617.8249

sub-category has highest discount

In [159...

```
sub_category = data.groupby('Sub-Category')['Discount'].sum()

print('Highest Discount on Sub-Category :', sub_category.idxmax())
print('Highest Discount Amount :', sub_category.max())
```

Highest Discount on Sub-Category : Binders

Highest Discount Amount : 567.0

Customer Analysis

• Who are the top 10 customers by sales?

In [172...

```
customers = data.groupby(['Customer ID', 'Customer Name'])['Sales'].sum().sort_v
print(customers)
```

Customer ID	Customer Name	
SM-20320	Sean Miller	25043.050
TC-20980	Tamara Chand	19052.218
RB-19360	Raymond Buch	15117.339
TA-21385	Tom Ashbrook	14595.620
AB-10105	Adrian Barton	14473.571
KL-16645	Ken Lonsdale	14175.229
SC-20095	Sanjit Chand	14142.334
HL-15040	Hunter Lopez	12873.298
SE-20110	Sanjit Engle	12209.438
CC-12370	Christopher Conant	12129.072

Name: Sales, dtype: float64

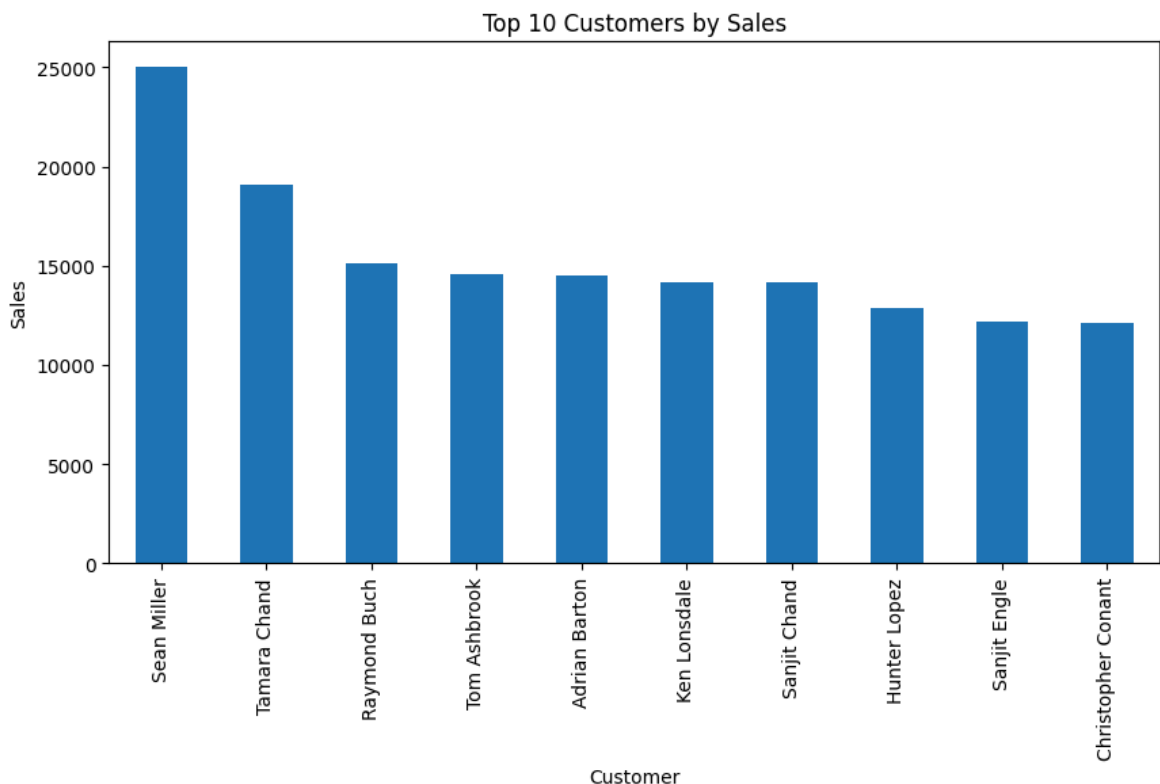
```
In [21]: top_customers = data.groupby('Customer Name')['Sales'].sum()\
        .sort_values(ascending=False)\
        .head(10)

plt.figure(figsize=(10,5))

top_customers.plot(kind='bar')

plt.title("Top 10 Customers by Sales")
plt.xlabel("Customer")
plt.ylabel("Sales")

plt.show()
```



The top 10 customers contribute a major share of overall sales revenue.

• Who are the most profitable customers?

```
In [177... customers = data.groupby(['Customer ID', 'Customer Name'])['Profit'].sum().sort_
print(customers)
```

```
Customer ID  Customer Name
TC-20980     Tamara Chand      8981.3239
Name: Profit, dtype: float64
```

Customer Tamara Chand (Customer ID: TC-20980) is the most profitable customer, generating a total profit of approximately 8981.32.

Among the customer segments (Consumer, Corporate, and Home Office), which segment buys the most?

```
In [180... segment = data.groupby('Segment')['Sales'].sum()

print('Highest Buying Segment :', segment.idxmax())
print('Sales Amount :', round(segment.max(), 2))
```

```
Highest Buying Segment : Consumer
Sales Amount : 1161401.34
```

• Which segment is most profitable?

```
In [181... segment = data.groupby('Segment')['Profit'].sum()

print('Highest Profitable Segment :', segment.idxmax())
print('Profit Amount :', round(segment.max(), 2))
```

```
Highest Profitable Segment : Consumer
Profit Amount : 134119.21
```

Product Analysis

• Which product has the highest sales?

```
In [185... product = data.groupby('Product Name')['Sales'].sum()

print('Highest Product Sales :', product.idxmax())
print('Amount :', round(product.max(), 2))
```

```
Highest Product Sales : Canon imageCLASS 2200 Advanced Copier
Amount : 61599.82
```

• Which product is giving loss despite high sales?

```
In [189... product = data.groupby('Product Name')[['Sales', 'Profit']].sum()

loss_product = product.sort_values(by='Profit').head(1)

print(loss_product)
```

Product Name	Sales	Profit
Cubify CubeX 3D Printer Double Head Print	11099.963	-8879.9704

• Which product is most frequently ordered?

```
In [197... product = data['Product Name'].value_counts()

print('Most Frequently Ordered Product :', product.idxmax())
print('Number of Orders :', product.max())
```

Most Frequently Ordered Product : Staple envelope
Number of Orders : 48

• Which products should be discontinued?

```
In [200... least_products = data['Product Name'].value_counts().sort_values().head(10)

print("Top 10 Least Ordered Products:")
print(least_products)
```

```
Product Name
Eldon Jumbo ProFile Portable File Boxes Graphite/Black    1
LG G2                                                        1
Eureka Hand Vacuum, Bagless                               1
Cisco 8961 IP Phone Charcoal                               1
Acco Glide Clips                                           1
Xerox 1901                                                  1
Memorex Mini Travel Drive 4 GB USB 2.0 Flash Drive        1
Jiffy Padded Mailers with Self-Seal Closure               1
Hewlett-Packard Deskjet 5550 Printer                      1
Avaya IP Phone 1140E VoIP phone                           1
Name: count, dtype: int64
```

Discount Analysis

• How does discount affect profit?

```
In [204... discount_effect = data.groupby('Discount')['Profit'].mean().sort_values(ascending=True)
print(discount_effect)
```

```
Discount
0.50    -310.703456
0.45    -226.646464
0.40    -111.927429
0.80    -101.796797
0.70     -95.874060
0.32     -88.560656
0.30     -45.679636
0.60     -43.077212
0.20      24.702572
0.15      27.288298
0.00      66.900292
0.10      96.055074
Name: Profit, dtype: float64
```

```
In [17]: plt.figure(figsize=(6,4))

discount_profit = data.groupby('Discount')['Profit'].mean()

plt.plot(discount_profit.index,
         discount_profit.values,
```

```

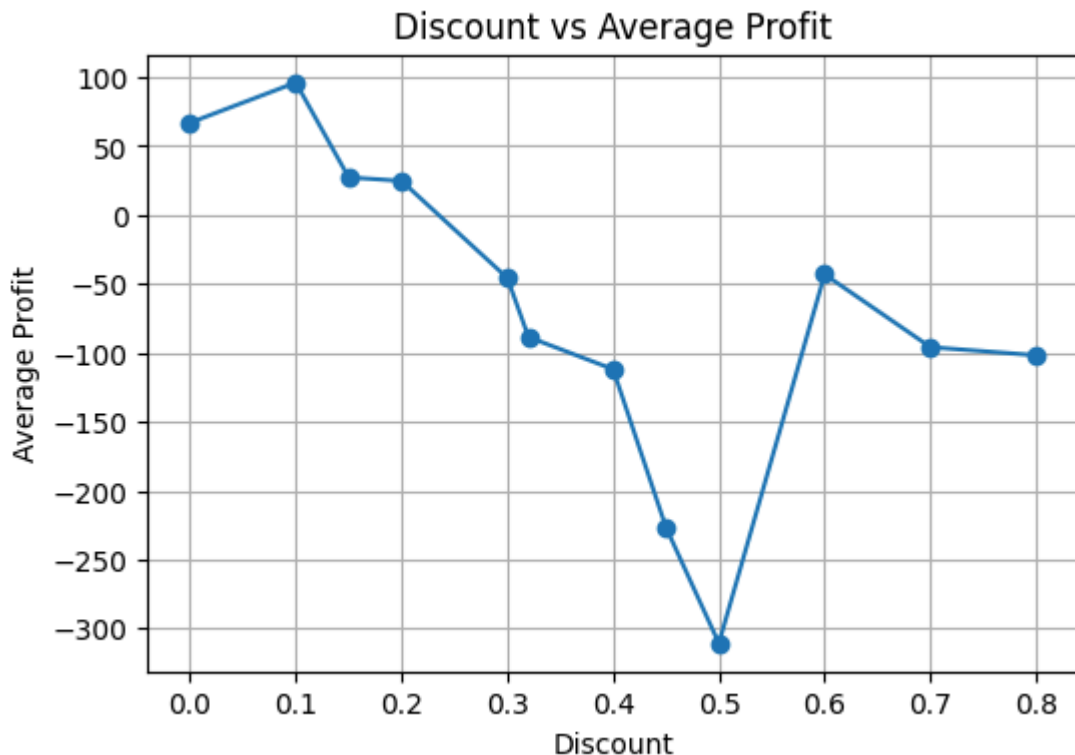
marker='o')

plt.title("Discount vs Average Profit")
plt.xlabel("Discount")
plt.ylabel("Average Profit")

plt.grid(True)

plt.show()

```



When discount crosses 40%, the company starts making heavy losses. The analysis shows that higher discounts generally reduce overall profit and can even lead to losses.

• Do high discounts always increase sales?

```

In [216... discount_affect = data.groupby('Discount')['Sales'].mean()
print(discount_affect)

```

```

Discount
0.00    226.742074
0.10    578.397351
0.15    529.971567
0.20    209.076940
0.30    454.742974
0.32    536.794770
0.40    565.134874
0.45    498.634000
0.50    892.705152
0.60     48.150000
0.70    97.177708
0.80    56.545853
Name: Sales, dtype: float64

```

“High discounts do not consistently lead to higher sales. The relationship is not strictly proportional, indicating that discounting alone is not a reliable strategy for increasing sales. A balanced approach considering product type and profitability is required.”

• Which discount range gives maximum profit?

```
In [222... discount_affect = data.groupby('Discount')['Profit'].mean()
print(discount_affect)
```

```
Discount
0.00    66.900292
0.10    96.055074
0.15    27.288298
0.20    24.702572
0.30   -45.679636
0.32   -88.560656
0.40  -111.927429
0.45  -226.646464
0.50  -310.703456
0.60   -43.077212
0.70   -95.874060
0.80  -101.796797
Name: Profit, dtype: float64
```

0%–10% discount → maximum profit zone 10%-30% discount → profit starts decreasing Above 30% → losses begin Best discount for maximum profit

```
In [224... discount_profit = data.groupby('Discount')['Profit'].mean()

best_discount = discount_profit.idxmax()
best_profit = discount_profit.max()

print("Best Discount:", best_discount)
print("Max Profit:", best_profit)
```

```
Best Discount: 0.1
Max Profit: 96.05507446808511
```

Small discounts (around 10%) are beneficial, but heavy discounts hurt profit significantly.

• Which products are affected negatively by discount?

```
In [230... product_discount_profit = data.groupby('Product Name')[['Discount', 'Profit']].m
print(product_discount_profit.sort_values('Profit'))
```

Product Name	Discount	Profit
Cubify CubeX 3D Printer Triple Head Print	0.500000	-3839.990400
Cubify CubeX 3D Printer Double Head Print	0.533333	-2959.990133
Cisco TelePresence System EX90 Videoconferencin...	0.500000	-1811.078400
Lexmark MX611dhe Monochrome Laser Printer	0.400000	-1147.493250
Zebra GK420t Direct Thermal/Thermal Transfer Pr...	0.700000	-938.280000
...	...	...
Zebra ZM400 Thermal Label Printer	0.000000	1671.768000
3D Systems Cube Printer, 2nd Generation, Magenta	0.000000	1858.985700
Ativa V4110MDD Micro-Cut Shredder	0.000000	1886.473050
Canon imageCLASS MF7460 Monochrome Digital Lase...	0.000000	1995.990000
Canon imageCLASS 2200 Advanced Copier	0.120000	5039.985600

[1850 rows x 2 columns]

```
In [231... loss_products = product_discount_profit[product_discount_profit['Profit'] < 0]
print(loss_products.sort_values('Profit'))
```

	Discount	Profit
Product Name		
Cubify CubeX 3D Printer Triple Head Print	0.500000	-3.839990e+03
Cubify CubeX 3D Printer Double Head Print	0.533333	-2.959990e+03
Cisco TelePresence System EX90 Videoconferencin...	0.500000	-1.811078e+03
Lexmark MX611dhe Monochrome Laser Printer	0.400000	-1.147493e+03
Zebra GK420t Direct Thermal/Thermal Transfer Pr...	0.700000	-9.382800e+02
...	...	...
Eldon Gobal File Keepers	0.066667	-1.009333e-01
Rubber Band Ball	0.050000	-7.480000e-02
Acco PRESSTEX Data Binder with Storage Hooks, D...	0.550000	-4.035000e-02
Premier Electric Letter Opener	0.075000	-8.881784e-16
Global Super Steno Chair	0.142857	-2.537653e-16

[301 rows x 2 columns]

Not all products should be discounted. Some products become loss-making when discounts are applied.

Shipping Analysis

• Which shipping mode is most used?

```
In [258... shipping_mode = data['Ship Mode'].value_counts()

print('Most shipping mode Used is :', shipping_mode.idxmax())
print('Count of shipping mode :', shipping_mode.max())
```

Most shipping mode Used is : Standard Class
Count of shipping mode : 5968

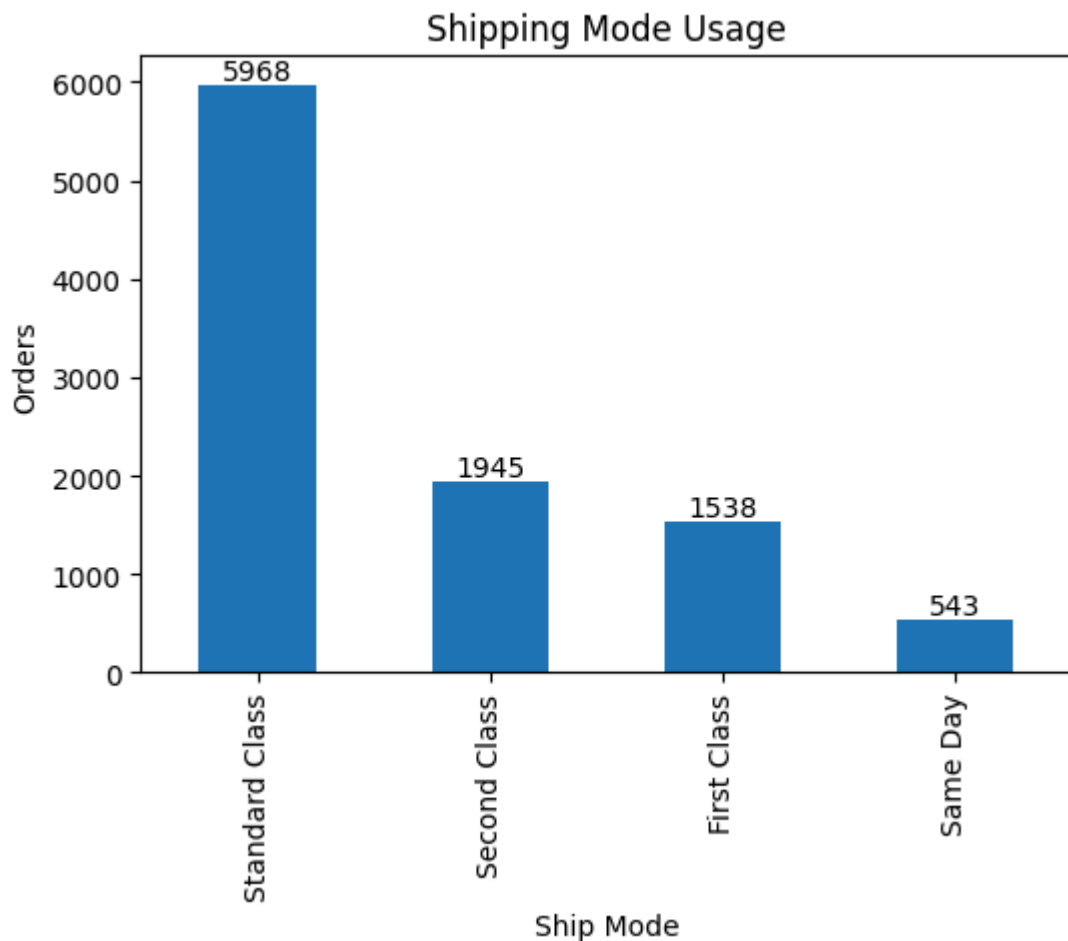
```
In [19]: plt.figure(figsize=(6,4))

ax = data['Ship Mode'].value_counts().plot(kind='bar')

plt.bar_label(ax.containers[0])

plt.title("Shipping Mode Usage")
plt.xlabel("Ship Mode")
plt.ylabel("Orders")

plt.show()
```



• Which shipping mode gives highest profit?

```
In [262... profit = data.groupby('Ship Mode')['Profit'].sum()

print("Highest Profit by Shipping Mode :", profit.idxmax())
print("Max Profit:", round(profit.max(),2))
```

Highest Profit by Shipping Mode : Standard Class
Max Profit: 164088.79

Time-Based Analysis

• Which month has highest sales?

Highest Sales Month

```
In [9]: data['Order Date'] = pd.to_datetime(data['Order Date'])
```

```
In [10]: data['Month'] = data['Order Date'].dt.month
```

```
In [12]: monthly_sales = data.groupby('Month')['Sales'].sum()

highest_month_sale = monthly_sales.idxmax()
highest_sale_amount = monthly_sales.max()

print("Highest Sales Month :", highest_month_sale)
print("Highest Month Sale Amount :", round(highest_sale_amount, 2))
```

Highest Sales Month : 11

Highest Month Sale Amount : 352461.07

```
In [316... fig, ax = plt.subplots(figsize=(8, 5))

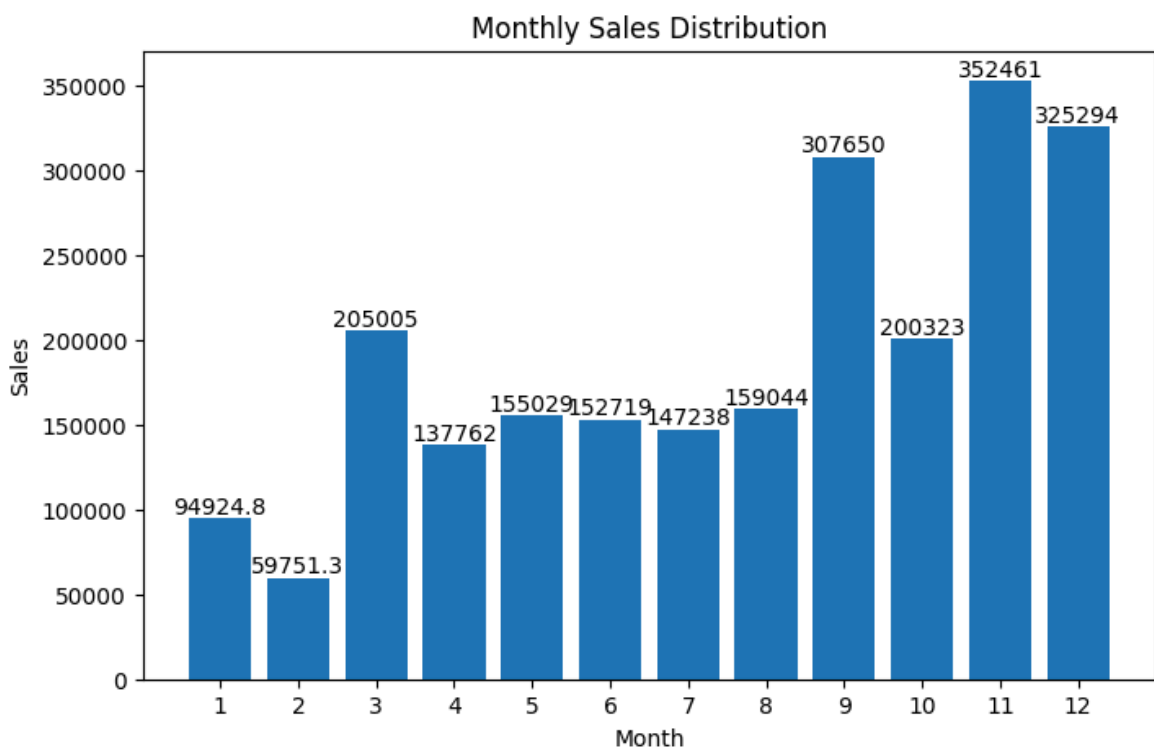
bars = ax.bar(monthly_sales.index, monthly_sales.values)

ax.bar_label(bars)

ax.set_xlabel("Month")
ax.set_ylabel("Sales")
ax.set_title("Monthly Sales Distribution")

ax.set_xticks(range(1, 13))

plt.show()
```



• Which year had maximum profit?

```
In [313... data['Order Date'] = pd.to_datetime(data['Order Date'])
```

```
In [314... data['year'] = data['Order Date'].dt.year
```

```
In [315... yearly_sales = data.groupby('year')['Profit'].sum().sort_values(ascending=False)
print(yearly_sales)
```

```
year
2017    93439.2696
Name: Profit, dtype: float64
```

The year 2017 recorded the highest sales of 93,439.27, indicating a strong upward trend in business performance compared to previous years.

• Is sales increasing or decreasing over time?

```
In [32]: year = [2014, 2015, 2016, 2017]
profits = [49543.9741, 61618.6037, 81795.1743, 93439.2696]

plt.figure(figsize=(4, 4))

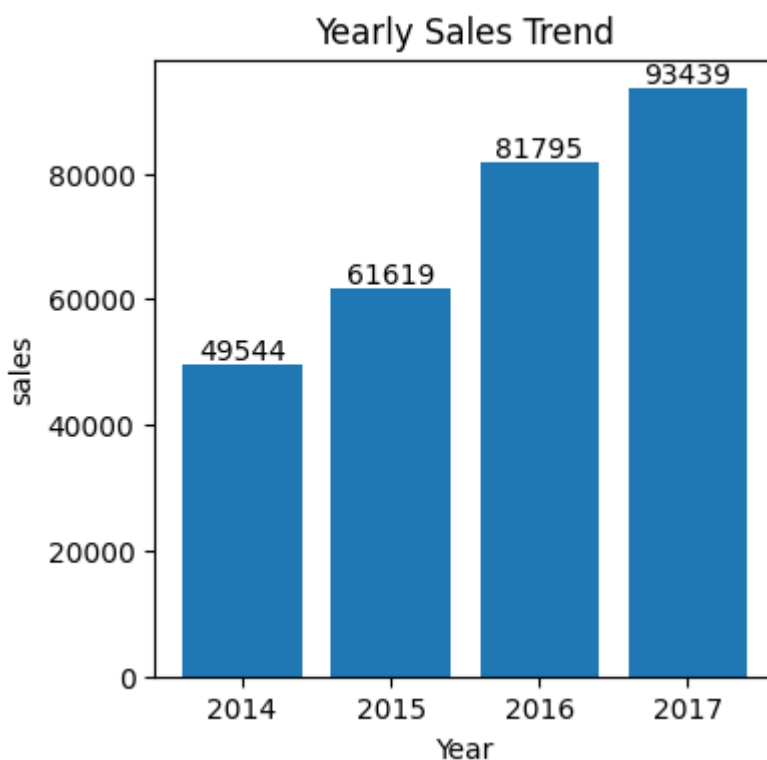
bars = plt.bar(year, profits)

plt.bar_label(bars, fmt="%.0f")

plt.xlabel("Year")
plt.ylabel("sales")
plt.title("Yearly Sales Trend")

plt.xticks(year)

plt.show()
```



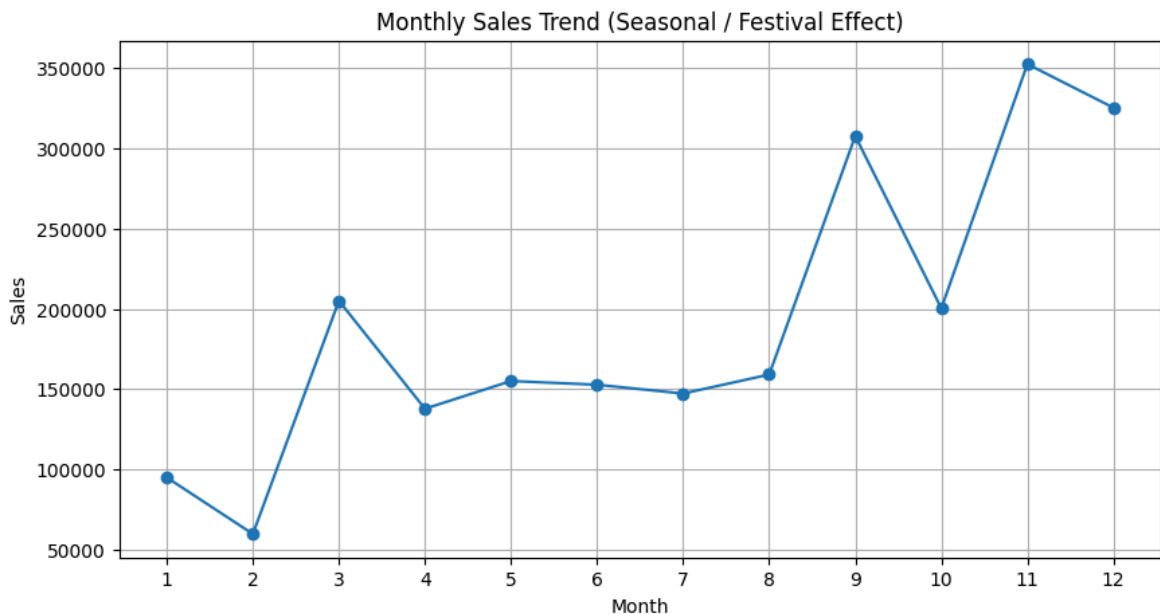
Sales are consistently increasing every year, showing a strong upward trend. From 2014 to 2015: growth of 24% From 2015 to 2016: growth of 33% From 2016 to 2017: growth of 14%

• Seasonal trend in sales (festival effect)

```
In [338... plt.figure(figsize=(10,5))
plt.plot(monthly_sales.index, monthly_sales.values, marker='o')

plt.xlabel("Month")
plt.ylabel("Sales")
plt.title("Monthly Sales Trend (Seasonal / Festival Effect)")
plt.xticks(range(1,13))
plt.grid(True)

plt.show()
```



Geography Insights

• Which region performs best overall?

```
In [350... region = data.groupby('Region')[['Sales', 'Profit']].sum().sort_values(by='Profit')
print(region)
```

	Sales	Profit
Region		
West	725457.8245	108418.4489
East	678781.2400	91522.7800
South	391721.9050	46749.4303
Central	501239.8908	39706.3625

```
In [349... region.head(1)
```

```
Out[349...
      Sales  Profit
Region
West  725457.8245  108418.4489
```

• Which state has highest sales but low profit?

```
In [353... state = data.groupby('State')[['Sales', 'Profit']].sum()

result = state[(state['Sales'] > state['Sales'].mean()) &
               (state['Profit'] < state['Profit'].mean())]

print(result.sort_values(by='Sales', ascending=False))
```

	Sales	Profit
State		
Texas	170188.0458	-25729.3563
Pennsylvania	116511.9140	-15559.9603
Florida	89473.7080	-3399.3017
Illinois	80166.1010	-12607.8870
Ohio	78258.1360	-16971.3766
North Carolina	55603.1640	-7490.9122

• Which city is most loss-making?

```
In [356... city = data.groupby('City')['Profit'].sum().sort_values(ascending=True)

print("Most Loss-Making City:")
print(city.head(1))
```

```
Most Loss-Making City:
City
Philadelphia    -13837.7674
Name: Profit, dtype: float64
```

Philadelphia is the most loss-making city with a total profit of -13,837.77, indicating inefficient profitability despite sales presence.

Business Insights

• What are the top 5 reasons for loss in this business?

The major causes of loss in the business are high discounting, inefficient shipping costs, specific loss-making cities, low-margin product categories, and poor pricing strategy. These factors result in situations where high sales do not translate into profit.

• Which products should company promote?

```
In [359... products = data.groupby('Product Name')[['Sales', 'Profit']].sum()

best_products = products.sort_values(by='Profit', ascending=False)

print(best_products.head(10))
```

	Sales	Profit
Product Name		
Canon imageCLASS 2200 Advanced Copier	61599.824	25199.9280
Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.384	7753.0390
Hewlett Packard LaserJet 3310 Copier	18839.686	6983.8836
Canon PC1060 Personal Laser Copier	11619.834	4570.9347
HP Designjet T520 Inkjet Large Format Printer -...	18374.895	4094.9766
Ativa V4110MDD Micro-Cut Shredder	7699.890	3772.9461
3D Systems Cube Printer, 2nd Generation, Magenta	14299.890	3717.9714
Plantronics Savi W720 Multi-Device Wireless Hea...	9367.290	3696.2820
Ibico EPK-21 Electric Binding System	15875.916	3345.2823
Zebra ZM400 Thermal Label Printer	6965.700	3343.5360

Products with high sales and positive profit are the best candidates for promotion, as they ensure revenue growth while maintaining profitability. Focusing on these products will maximize business efficiency and return on marketing investment.

• Where should company reduce discount?

```
In [361... discount_area = data.groupby('State')[['Discount', 'Profit', 'Sales']].mean()

print(discount_area.sort_values(by='Profit'))
```

State	Discount	Profit	Sales
Ohio	0.324947	-36.186304	166.861697
Colorado	0.316484	-35.867351	176.418231
North Carolina	0.283534	-30.083985	223.305880
Tennessee	0.291257	-29.189583	167.551219
Pennsylvania	0.328620	-26.507598	198.487077
Texas	0.370193	-26.121174	172.779742
Illinois	0.390041	-25.625787	162.939230
Arizona	0.303571	-15.303235	157.508933
Oregon	0.288710	-9.600569	140.573790
Florida	0.299347	-8.875461	233.612815
New Mexico	0.059459	31.273408	129.284378
North Dakota	0.000000	32.878529	131.415714
South Dakota	0.000000	32.902358	109.630000
Kansas	0.000000	34.851813	121.429583
California	0.072764	38.171608	228.729451
Idaho	0.085714	39.367767	208.689810
Iowa	0.000000	39.460397	152.658667
South Carolina	0.000000	42.120395	201.945476
Connecticut	0.007317	42.823071	163.223866
West Virginia	0.075000	46.480400	302.456000
Utah	0.060377	48.047802	211.699170
Massachusetts	0.015556	50.262975	212.106919
Louisiana	0.000000	52.288150	219.453095
Nebraska	0.000000	53.607742	196.445526
Maine	0.000000	56.810775	158.816250
Mississippi	0.000000	59.867475	203.232830
New Hampshire	0.011111	63.203807	270.093481
New York	0.055319	65.637011	275.599531
Washington	0.064032	66.013146	273.994605
Arkansas	0.000000	66.811452	194.635500
Maryland	0.005714	66.963608	225.766886
Oklahoma	0.000000	73.544788	298.233182
New Jersey	0.004615	75.176260	275.110092
Wisconsin	0.000000	76.380004	291.951000
Kentucky	0.000000	80.573357	263.250000
Virginia	0.000000	83.026564	315.342500
Nevada	0.061538	85.045279	428.951333
Georgia	0.000000	88.315453	266.825217
Alabama	0.000000	94.865989	319.846557
Michigan	0.007059	95.934069	299.096525
Missouri	0.000000	97.518341	336.441667
Wyoming	0.200000	100.196000	1603.136000
Delaware	0.006250	103.930988	285.948635
District of Columbia	0.000000	105.958930	286.502000
Minnesota	0.000000	121.608847	335.541011
Montana	0.066667	122.221900	372.623467
Indiana	0.000000	123.375411	359.431946
Rhode Island	0.021429	130.100523	404.070643
Vermont	0.000000	204.088936	811.760909

```
In [362... risk_states = discount_area[(discount_area['Discount'] > discount_area['Discount']
                                (discount_area['Profit'] < 0)]

print(risk_states.sort_values(by='Profit'))
```

	Discount	Profit	Sales
State			
Ohio	0.324947	-36.186304	166.861697
Colorado	0.316484	-35.867351	176.418231
North Carolina	0.283534	-30.083985	223.305880
Tennessee	0.291257	-29.189583	167.551219
Pennsylvania	0.328620	-26.507598	198.487077
Texas	0.370193	-26.121174	172.779742
Illinois	0.390041	-25.625787	162.939230
Arizona	0.303571	-15.303235	157.508933
Oregon	0.288710	-9.600569	140.573790
Florida	0.299347	-8.875461	233.612815

Discounts should be reduced in states where high discount rates are associated with low or negative profit, as excessive discounting is directly harming profitability.

• Which region should company expand?

```
In [364... region = data.groupby('Region')[['Sales', 'Profit']].sum()

region = region.sort_values(by='Profit', ascending=False)

print(region)
```

	Sales	Profit
Region		
West	725457.8245	108418.4489
East	678781.2400	91522.7800
South	391721.9050	46749.4303
Central	501239.8908	39706.3625

```
In [365... region['Profit Margin'] = region['Profit'] / region['Sales'] * 100

print(region.sort_values(by='Profit Margin', ascending=False))
```

	Sales	Profit	Profit Margin
Region			
West	725457.8245	108418.4489	14.944831
East	678781.2400	91522.7800	13.483399
South	391721.9050	46749.4303	11.934342
Central	501239.8908	39706.3625	7.921629

The company should expand in regions with high profit and strong profit margins, as these indicate efficient operations and strong market demand.

• What is the most profitable strategy combination?

```
In [366... combo = data.groupby(['Segment', 'Region'])[['Sales', 'Profit']].sum()

best_combo = combo.sort_values(by='Profit', ascending=False)

print(best_combo.head(10))
```

		Sales	Profit
Segment	Region		
	Consumer		
	West	362880.7730	57450.6040
	East	350908.1670	41190.9843
Corporate	West	225855.2745	34437.4299
Consumer	South	195580.9710	26913.5728
Home Office	East	127463.7260	26709.2168
Corporate	East	200409.3470	23622.5789
	Central	157995.8128	18703.9020
Home Office	West	136721.7770	16530.4150
Corporate	South	121885.9325	15215.2232
Home Office	Central	91212.6440	12438.4124

```
In [367... cat_combo = data.groupby(['Category', 'Sub-Category'])[['Sales', 'Profit']].sum()

best_cat = cat_combo.sort_values(by='Profit', ascending=False)

print(best_cat.head(10))
```

		Sales	Profit
Category	Sub-Category		
	Technology		
	Copiers	149528.030	55617.8249
	Phones	330007.054	44515.7306
	Accessories	167380.318	41936.6357
Office Supplies	Paper	78479.206	34053.5693
	Binders	203412.733	30221.7633
Furniture	Chairs	328449.103	26590.1663
Office Supplies	Storage	223843.608	21278.8264
	Appliances	107532.161	18138.0054
Furniture	Furnishings	91705.164	13059.1436
Office Supplies	Envelopes	16476.402	6964.1767

```
In [368... ship = data.groupby('Ship Mode')[['Sales', 'Profit']].sum()

print(ship.sort_values(by='Profit', ascending=False))
```

		Sales	Profit
Ship Mode			
Standard Class	1.358216e+06	164088.7875	
Second Class	4.591936e+05	57446.6354	
First Class	3.514284e+05	48969.8399	
Same Day	1.283631e+05	15891.7589	

```
In [369... discount_strategy = data.groupby('Discount')[['Sales', 'Profit']].sum()

print(discount_strategy.sort_values(by='Profit', ascending=False))
```

		Sales	Profit
Discount			
0.00	1.087908e+06	320987.6032	
0.20	7.645944e+05	90337.3060	
0.10	5.436935e+04	9029.1770	
0.15	2.755852e+04	1418.9915	
0.32	1.449346e+04	-2391.1377	
0.45	5.484974e+03	-2493.1111	
0.60	6.644700e+03	-5944.6552	
0.30	1.032267e+05	-10369.2774	
0.50	5.891854e+04	-20506.4281	
0.40	1.164178e+05	-23057.0504	
0.80	1.696376e+04	-30539.0392	
0.70	4.062028e+04	-40075.3569	

The most profitable strategy is achieved by combining high-performing segments and regions with high-margin product categories, efficient shipping modes, and controlled discount levels to maximize profit.

KPI Summary

```
In [27]: # Total Sales
total_sales = round(data['Sales'].sum(), 2)

# Total Profit
total_profit = round(data['Profit'].sum(), 2)

# Profit Margin
profit_margin = round((total_profit / total_sales) * 100, 2)

# Total Orders
total_orders = data['Order ID'].nunique()

# Total Customers
total_customers = data['Customer ID'].nunique()

# Best Region (Highest Profit)
best_region = data.groupby('Region')['Profit'].sum().idxmax()

# Worst State (Lowest Profit)
worst_state = data.groupby('State')['Profit'].sum().idxmin()

print("==== KPI SUMMARY =====")

print("Total Sales :", total_sales)
print("Total Profit :", total_profit)
print("Profit Margin :", profit_margin, "%")
print("Total Orders :", total_orders)
print("Total Customers :", total_customers)
print("Best Region :", best_region)
print("Worst State :", worst_state)
```

```
==== KPI SUMMARY =====
Total Sales : 2297200.86
Total Profit : 286397.02
Profit Margin : 12.47 %
Total Orders : 5009
Total Customers : 793
Best Region : West
Worst State : Texas
```

Key Findings: The business is overall profitable. Technology category generated strong profits. Heavy discounts reduced profitability significantly. Philadelphia was the most loss-making city. Standard Class shipping mode generated the highest profit. Sales increased strongly during November and December. Some products had high sales but negative profits. Important Observation: Higher sales do not always mean higher profits. Discounts and operational costs strongly affect profitability.

Recommendations

Based on the analysis of the Superstore dataset, several strategic recommendations can help improve overall business performance and profitability.

1. Reduce Excessive Discounts

High discounts were one of the major reasons for negative profits in several products and regions. The company should carefully control discount rates, especially in low-profit areas.

2. Focus on High-Profit Categories

The Technology category generated strong profits compared to other categories. The business should increase promotion, inventory, and marketing efforts for high-performing products.

3. Improve Performance in Loss-Making Cities

Cities such as Philadelphia showed significant losses. The company should analyze operational costs, pricing strategies, and customer demand in these areas.

4. Optimize Shipping Strategy

Standard Class shipping mode generated the highest profit. The business should optimize shipping operations to reduce delivery costs while maintaining customer satisfaction.

5. Increase Marketing During Peak Sales Months

Sales performance was highest during November and December. The company should launch seasonal campaigns and promotional offers during these months to maximize revenue.

6. Monitor Products with High Sales but Low Profit

Some products generated strong sales but low or negative profit margins. These products should be re-evaluated for pricing, discounts, and operational costs.

Based on the analysis, the following business recommendations were provided:

Conclusion

This project demonstrates practical business analysis using Python and data visualization techniques. The analysis successfully identified important sales trends, profit patterns, operational issues, and business opportunities.